

Lymphogranuloma Venereum

(MONDO:0005834): A Comprehensive Disease Characteristics Report

Target disease: Lymphogranuloma Venereum · **MONDO:0005834** · Category: Infectious Disease **Report type:** Disease-level synthesis from primary literature and clinical guidelines (no patient-level data files were provided; all evidence is aggregated/literature-derived). **Date:** 2026-09-25

Summary

Lymphogranuloma venereum (LGV) is an invasive, systemic sexually transmitted infection caused by the **L-biovar of *Chlamydia trachomatis*** (serovars L1, L2, L2b, and L3). Unlike the non-invasive "trachoma" biovar that causes ocular and uncomplicated urogenital chlamydial infection, the LGV strains are lymphotropic and macrophage-tropic, enabling them to disseminate from the mucosal inoculation site into the regional lymphatics. This distinction — genomic and biological rather than genetic-in-the-host — is the central determinant of the disease. LGV is **not a human Mendelian disease**; there are no causal human genes, pathogenic germline variants, or heritable susceptibility loci. The "genotype" that matters is the pathogen's, defined by *ompA/pmpH* genovar and the cryptic virulence plasmid.

Clinically, LGV classically unfolds in three stages: (1) a transient primary papule or ulcer at the site of inoculation; (2) secondary regional (usually unilateral) inguinal lymphadenopathy with buboes — or, in the modern epidemic among men who have sex with men (MSM), a hemorrhagic **anorectal proctitis**; and (3) a tertiary fibrotic "genito-anorectal syndrome" producing rectal strictures, fistulae, and genital elephantiasis if left untreated. Since 2003, LGV has re-emerged across Europe, North America, Australia, and New Zealand as an epidemic among predominantly **HIV-positive MSM**, driven overwhelmingly by the **L2b** genovar, with proctitis now the dominant presentation and roughly a quarter of anorectal infections asymptomatic.

Diagnosis requires a two-step laboratory approach: detection of *C. trachomatis* by nucleic acid amplification test (NAAT) followed by LGV-discriminatory genotyping (*pmpH/ompA* PCR). First-line therapy is **doxycycline 100 mg twice daily for 21 days**, which achieves a pooled microbial cure rate of **98.5%** in rectal LGV. Prevention rests on condoms, partner treatment, screening, and increasingly **doxycycline post-exposure prophylaxis (doxy-PEP)**; no vaccine exists. The pathophysiology is best understood as a causal chain running from Type III secretion system (T3SS)-mediated epithelial invasion, through productive replication in macrophages and lymphatic dissemination, to an IFN- γ /Th1-driven chronic inflammatory response in which matrix metalloproteinase (MMP)-mediated tissue remodeling produces the characteristic scarring and fibrosis. This report synthesizes 15 evidence-backed findings drawn from 37 reviewed papers, organized against the 15-section disease-characteristics template.

Key Findings

1. Disease Information

LGV is an invasive systemic infection caused by *C. trachomatis* serovars L1, L2, and L3 (with L2b predominating in current outbreaks), following a three-stage clinical course. As the etiologic review by Ceovic and Gulin states: *"The etiological agent of LGV is Chlamydia trachomatis serotypes L1, L2 and L3, and current outbreaks are mostly sustained by L2b type. The clinical course can be classically divided into three stages: an initial papule, which may ulcerate at the site of inoculation, followed by regional lymphadenopathy (second stage, generally unilateral). In the tertiary stage, lymphatic obstruction, with elephantiasis of genitalia, and rectal involvement can lead to the formation of strictures and fistulae that may require surgical treatment"* (PMID: 24518282).

Key identifiers and synonyms:

Resource	Identifier / Term
MONDO	MONDO:0005834
ICD-10	A55 (Chlamydial lymphogranuloma [venereum])
ICD-11	1A75 (Lymphogranuloma venereum)
MeSH	Lymphogranuloma Venereum (D008219)
SNOMED CT	186946009
Synonyms	Lymphogranuloma inguinale; Durand-Nicolas-Favre disease; climatic bubo; tropical bubo; poradenitis; lymphopathia venereum; "anorectal syndrome" (modern)

The information is derived from **aggregated disease-level resources** — clinical reviews, surveillance datasets, guideline documents, and case series — rather than individual EHR records. (Sources: [PMID: 24518282](#); [PMID: 39915233](#))

2. Etiology

Causal factor: infectious. LGV is caused entirely by infection with the invasive L-biovar of *C. trachomatis*. There is no genetic (host) etiology, no heritable susceptibility variant, no modifier gene, and no gene–environment interaction in the classical human-genetics sense. The determinism lies in the pathogen: *C. trachomatis* comprises **two biovariants** — the **non-invasive "trachoma" biovar** (ocular and genital serovars A–K) **and the invasive "lymphogranuloma venereum" strains** — and *"the plasmid has been linked to chlamydial virulence"* ([PMID: 19460133](#)).

Risk factors (environmental/behavioral): - Being a man who has sex with men (MSM) — 93.8% of a 161-isolate Spanish series were MSM ([PMID: 39053939](#)) - HIV coinfection — $\geq 43.5\%$ in the same series; $\sim 74\%$ of UK diagnoses in 2013 ([PMID: 39053939](#); [PMID: 32762828](#)) - Condomless anal sex, multiple partners, concurrent STIs, chemsex - Receptive anal intercourse (anatomical route for the dominant anorectal syndrome)

Protective factors: condom use (imperfect), partner notification and treatment, and doxycycline post-exposure prophylaxis (see Section 13). There are **no known genetic protective variants** because the host genome does not determine disease.

3. Phenotypes

LGV presents as two overlapping clinical syndromes: the **classic inguinal syndrome** and the modern **anorectal proctitis syndrome**.

Phenotype	Type	Suggested HPO term	Frequency / notes
Genital papule/ulcer (primary)	Physical manifestation	HP:0200037 (Genital ulceration)	Often transient/unnoticed
Inguinal lymphadenopathy / buboes	Clinical sign	HP:0002716 (Lymphadenopathy)	Classically unilateral
Proctitis (anorectal pain, tenesmus, discharge, bleeding)	Symptom/sign	HP:0002607 (rectal); proctitis	Proctitis in 73.3% of symptomatic MSM
Anorectal bleeding	Symptom	HP:0002573 (Hematochezia)	Common in proctitis
Constipation / tenesmus	Symptom	HP:0002019 (Constipation)	Common
Rectal stricture (tertiary)	Physical manifestation	Rectal fibrosis/stricture	Late complication
Fistula formation (tertiary)	Physical manifestation	Anal fistula	Late complication
Genital elephantiasis (tertiary)	Physical manifestation	HP:0001004 (Lymphedema)	Late complication
Asymptomatic carriage	—	—	~25% of anorectal infections

The 2019 European guideline notes: *"Among MSM, about 25% of the anorectal LGV infections are asymptomatic"* (PMID: 31243838). Proctitis symptoms are detailed in the 2021 European proctitis guideline: *"The symptoms of proctitis include anorectal itching, pain, tenesmus, bleeding, constipation and discharge in and around the anal canal. The majority of rectal chlamydia and gonococcal infections are asymptomatic and can only be detected by laboratory tests"* (PMID: 34057249). The modern picture is dominated by *"progressive ulcerative proctitis, the so called anorectal syndrome"* (PMID: 24518282).

Onset/severity/progression: adult-onset; severity ranges mild-to-severe and is **serovar-dependent** (L1 milder than L2, see Finding 12); progression is from acute proctitis to chronic fibrosis if untreated. **Quality-of-life impact** is substantial in symptomatic proctitis (pain, tenesmus, bleeding) and severe in the tertiary stage (disfiguring elephantiasis, strictures requiring surgery), but disease-specific validated QoL instruments (EQ-5D/SF-36) have not been applied — a knowledge gap.

4. Genetic/Molecular Information

Not applicable to the human host. LGV has **no causal human genes, pathogenic germline/somatic variants, modifier genes, host epigenetic lesions, or chromosomal abnormalities.** The relevant molecular information belongs to the pathogen:

- **Pathogen biovar/genovar:** the invasive LGV biovar is genomically distinct from the trachoma biovar ([PMID: 19460133](#)).
- **Cryptic plasmid:** the ~7.5-kb *C. trachomatis* plasmid is linked to virulence and is a key diagnostic target ([PMID: 19460133](#)).
- **Genotyping markers:** *ompA* (encoding the major outer membrane protein, MOMP) and *pmpH* (polymorphic membrane protein H) discriminate LGV from non-LGV strains ([PMID: 22517888](#)).
- **Ongoing evolution:** long-term reference-laboratory surveillance (Portuguese NRL, 1,188 LGV isolates) has identified newly emerging genovars, including a recently emerging **L1-like variant** ([PMID: 39915233](#)).

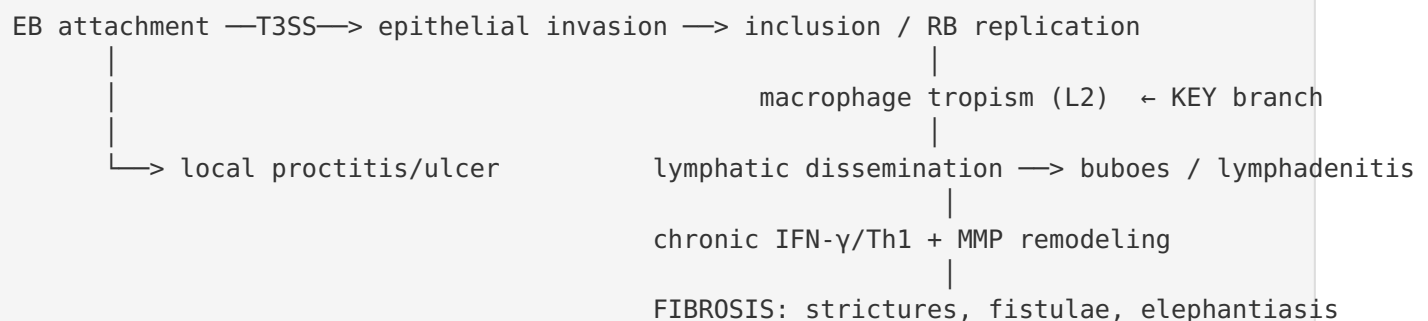
5. Environmental Information

Infectious agent: *Chlamydia trachomatis* (NCBI Taxonomy ID 813), L-biovar (serovars L1, L2, L2b, L3). An obligate intracellular Gram-negative bacterium. The relevant "environmental" factors are behavioral (sexual network exposure among MSM, chemsex, condomless anal sex) rather than chemical/toxic. No toxin, radiation, pollutant, occupational exposure, or dietary factor is causally implicated. Transmission is sexual (skin/mucosa contact), with the anorectal mucosa the dominant portal in the current epidemic. (Sources: [PMID: 24518282](#); [PMID: 39053939](#))

6. Mechanism / Pathophysiology

Ordered causal chain (initiating infection → clinical manifestation):

1. Sexual exposure deposits infectious **elementary bodies (EBs)** of the L-biovar onto genital or anorectal mucosal epithelium → **leads to** attachment and entry.
2. *C. trachomatis* delivers T3SS effectors into the host epithelial cell → **facilitates** invasion and formation of the membrane-bound inclusion (loss of even a single effector such as CT622 strongly reduces invasion). (*demonstrated in vitro*)
3. Inside the inclusion, EBs differentiate to metabolically active **reticulate bodies (RBs)**, replicate, and re-differentiate to EBs (biphasic developmental cycle) → **results in** host-cell egress and local spread. (*demonstrated*)
4. Unlike non-disseminating urogenital serovars, the disseminating **serovar L2 replicates productively in monocyte-derived macrophages** → **enables** the bacteria to survive within phagocytes and be carried into regional lymphatics. (*demonstrated in vitro; the key branch point that makes LGV invasive*)
5. Infected/activated cells release a distinctive cytokine profile (greater IL-8, IL-11; differential TNF- α ; IDO upregulation) → **drives** neutrophil and macrophage recruitment and a pro-dissemination inflammatory milieu. (*demonstrated in vitro*)
6. Lymphatic spread to draining nodes → **produces** lymphangitis, lymphadenitis, and buboes (inguinal syndrome) or hemorrhagic proctitis (anorectal syndrome). (*clinical*)
7. A sustained **IFN- γ -dependent Th1** response controls the organism but, when chronic, together with altered **matrix metalloproteinase (MMP)** activity, injures tissue → **causes** fibrosis, strictures, fistulae, and elephantiasis. (*extrapolated from trachoma, the best-characterized chlamydial scarring disease*)



Supporting evidence. The T3SS mechanism: "*C. trachomatis* delivers an arsenal of virulence factors into the eukaryotic cell via a type 3 secretion system (T3SS) that facilitates invasion, manipulation of host vesicular trafficking, subversion of host defense mechanisms and promotes bacteria egress at the conclusion of the developmental cycle"; and "All chlamydiae are obligate intracellular bacteria that replicate within a membrane-bound vacuole termed the inclusion" (PMID: 33512479). Macrophage tropism: "In contrast to the abortive replication of *C. trachomatis* in monocytes, monocyte-derived macrophages permitted replication as indicated by one-step growth experiments and TEM" (PMID: 3759241). Differential cytokine induction: "Infection of HeLa cells with *C. trachomatis* E or L2 induced a strong and similar PMN chemotactic response, but larger amounts of interleukin (IL)-8 and IL-11 were released after infection with serovar L2" (PMID: 11207588), with additional differential TNF- α /IDO signaling (PMID: 12011019). Scarring mechanism: "An increasing number of studies indicate that innate immune responses arising from the epithelium and other innate immune cells, along with changes in matrix metalloproteinase activity, are important in the development of tissue damage and scarring," and "The resolution of *Ct* infection in animal models is IFN γ -dependent, involving Th1 cells" (PMID: 23457650).

Suggested ontology terms: GO:0009405 (pathogenesis); GO:0051701 (biological process involved in interaction with host); GO:0030260 (entry into host cell); GO:0006954 (inflammatory response); GO:0042088 (T-helper 1 type immune response); GO:0030574 (collagen catabolic process / MMP activity). Cell types: CL:0000235 (macrophage); CL:0000775 (neutrophil); CL:0000066 (epithelial cell); CL:0000545 (T-helper 1 cell). Chemical entities: CHEBI:15551 (prostaglandin-related mediators, illustrative); CHEBI:doxycycline (CHEBI:50845).

7. Anatomical Structures Affected

Level	Structure	UBERON / CL / GO term
Primary organ	Rectum / anorectal mucosa (modern)	UBERON:0001052 (rectum)
Primary organ	External genitalia / inguinal skin (classic)	UBERON:0000079 (male reproductive system)
Primary organ	Regional lymph nodes (inguinal, iliac, perirectal)	UBERON:0000029 (lymph node)
Secondary	Lymphatic vessels	UBERON:0001473 (lymphatic vessel)
Body system	Lymphatic/immune; lower digestive tract; genital system	—
Tissue	Mucosal epithelium; connective tissue (fibrosis)	UBERON:0000483 (epithelium)
Cells targeted	Epithelial cells; macrophages/monocytes	CL:0000066; CL:0000235
Subcellular	Membrane-bound inclusion (pathogen vacuole)	GO:0030430 (host cell cytoplasm)

Lateralization: the classic inguinal syndrome is characteristically **unilateral** ([PMID: 24518282](#)); anorectal disease is midline/rectal.

8. Temporal Development

- **Onset:** adult-onset, acute-to-subacute. Primary lesion appears days to weeks after exposure; secondary stage weeks later.
- **Stages:** primary (papule/ulcer) → secondary (lymphadenitis/buboes or proctitis) → tertiary (fibrosis).
- **Progression:** untreated disease is slowly progressive toward fibrosis; treated disease resolves promptly. In a Polish case series the **median diagnostic delay was 6 months** (range 2 weeks–7 months), with patients consulting a median of 3 physicians before diagnosis, and colonoscopy findings *"initially suggesting inflammatory bowel disease or malignancy"* ([PMID: 42151834](#)).

- **Duration/remission:** self-limited/curable with antibiotics; tertiary damage may be irreversible. Delayed treatment (≥ 6 months) is associated with **persistent anorectal symptoms** despite therapy (PMID: 42151834).
- **Critical period for intervention:** early antibiotic therapy (before fibrosis) is the window of opportunity.

9. Inheritance and Population

Inheritance: none — this is an infectious disease with **no inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency** in the human genetic sense.

Epidemiology and demographics: - **Predominant population:** HIV-positive MSM. In Spain (2018–2019, 161 isolates): *"Most of the 161 LGV isolates (93.8%) were detected in men who have sex with men (MSM). At least 43.5% of the patients presented with HIV coinfection and 53.4% were symptomatic, with proctitis being the most prevalent symptom (73.3%)"* (PMID: 39053939). - **Geographic:** re-emergent since 2003 across Europe, Australia, New Zealand, the US, and Canada (PMID: 24518282). Historically endemic in tropical/subtropical regions. - **Sex ratio:** heavily male-predominant in the current epidemic; heterosexual LGV is *"extremely rare"* in Europe with no evidence of heterosexual transmission (PMID: 31243838). - **Anatomical distribution:** genital:anorectal ratio $\approx 1:15$ among MSM; L2b and L2 predominate (PMID: 31243838). - **Surveillance trend (UK):** annual diagnoses rose from 28 (2004) to 904 (2016), then fell to 641 (2017); test positivity halved from 14.8% (2015) to 7.3% (2018); the HIV-positive share of diagnoses fell from 74% (2013) to 48% (2018) (PMID: 32762828).

10. Diagnostics

Two-step laboratory algorithm: (1) detect *C. trachomatis* by NAAT (e.g., Aptima Combo 2) on the relevant anatomical site; (2) confirm LGV by genotyping. In a Finnish diagnostic study: *"Altogether 140 C trachomatis NAAT-positive rectal and pharyngeal samples were genotyped by pmpH and ompA real-time PCR. Of the 140 NAAT-positive rectal and pharyngeal specimens, 114 (81%) were successfully typed by pmpH PCR"* — with LGV (mostly L2b) found mainly in rectal samples (PMID: 22517888). The 2019 European guideline defines diagnosis as a *C. trachomatis*-positive NAAT confirmed by an LGV-discriminatory NAAT (PMID: 31243838).

Diagnosis is often delayed and requires clinical suspicion: *"Diagnosis is often delayed, requires a high index of clinical suspicion and must rely on the use of nucleic acid amplification tests"* (PMID: 24518282).

Endoscopy/histopathology: LGV proctitis endoscopically and histologically **mimics inflammatory bowel disease (IBD)** and malignancy. In HIV-infected men: *"four cases of chlamydial proctitis in HIV-infected individuals, who had different clinical presentations but very similar endoscopic and histopathologic features, as well as prompt and complete response to therapy"* (PMID: 16721218).

Differential diagnosis: IBD (Crohn's proctitis), rectal malignancy, other ulcerative STIs (syphilis, HSV, chancroid, donovanosis), and non-LGV chlamydial/gonococcal proctitis. **Serology** (complement fixation) is historical and non-specific. **Genetic testing / omics-based diagnostics / newborn or carrier screening: not applicable.**

11. Outcome / Prognosis

Prognosis is excellent with timely antibiotic therapy and poor-to-morbid if untreated. Early treatment yields prompt, complete resolution (PMID: 16721218). Untreated tertiary disease causes disfiguring, potentially irreversible complications — rectal strictures, fistulae, chronic proctocolitis, and genital elephantiasis — that may require surgery (PMID: 24518282).

- **Mortality:** LGV is essentially non-fatal with treatment; deaths are exceptional and relate to untreated complications or secondary infection. No formal survival statistics apply.
- **Morbidity:** substantial in the tertiary stage (disability from strictures/lymphedema); moderate in symptomatic proctitis.
- **Prognostic factor:** diagnostic delay. Patients with delays ≥ 6 months reported persistent anorectal symptoms despite standard therapy (PMID: 42151834).
- **Recovery:** high with early treatment; incomplete once fibrosis is established.

12. Treatment

First-line pharmacotherapy: doxycycline 100 mg orally twice daily for 21 days (NCIT: C312 Doxycycline; ATC J01AA02; CHEBI:50845). This prolonged tetracycline course reflects the invasive, systemic nature of LGV versus the 7-day/single-dose regimens used for uncomplicated urogenital chlamydia. In a Polish case series, *"All received doxycycline 100 mg twice daily for 21 days"* (PMID: 42151834). The 2019 European guideline recommends the same regimen — *"Doxycycline 100 mg twice a day orally for 21 days is the recommended treatment for LGV. This same treatment is recommended also in asymptomatic patients and contacts of LGV patients. If another regimen is used, a test of cure (TOC) must be performed"* (PMID: 31243838).

Efficacy (quantitative): A systematic review and meta-analysis of 9 studies (282 MSM with rectal LGV) found: *"The fixed-effects pooled efficacy for doxycycline was 98.5% (95% CI 96.3%-100%, $I^2 = 0\%$; $p = 0.993$). Doxycycline at 100 mg twice daily for 21 days demonstrated a high microbial cure rate"* (PMID: 27513890).

Regimen	Dose / duration	Role	NCIT / evidence
Doxycycline	100 mg BID × 21 days	First-line	NCIT C312; 98.5% cure (PMID: 27513890)
Azithromycin	1 g weekly × 3 weeks	Alternative (requires TOC)	NCIT C1264; guideline (PMID: 31243838)
Erythromycin	500 mg QID × 21 days	Alternative (e.g., pregnancy)	NCIT C609; guideline

Surgical/interventional: drainage/aspiration of fluctuant buboes; surgical repair of strictures/fistulae in tertiary disease. **Supportive care** for proctitis symptoms. **Pharmacogenomics, gene/cell/RNA therapy, targeted/immunotherapy: not applicable.** Partners and asymptomatic contacts are treated (PMID: 31243838).

13. Prevention

- **Primary prevention:** condom use (imperfect protection), reduction in partner numbers, partner notification and empiric treatment of contacts (PMID: 31243838); condoms *"do not guarantee protection"* (PMID: 34057249).
- **Doxycycline post-exposure prophylaxis (doxy-PEP):** doxycycline 200 mg after condomless sex substantially reduces chlamydia (and syphilis) incidence among high-risk MSM and is being rolled out (England, 2025). *"Doxycycline post-exposure prophylaxis (doxy-PEP) has been shown to reduce the incidence of chlamydia and syphilis among men who have sex with men (MSM) at high risk of sexually transmitted infections (STIs)"* (PMID: 42565264); independently, *"Promising prevention innovations include doxycycline post-exposure prophylaxis, which substantially reduces chlamydia and syphilis incidence"* (PMID: 42538441). Population modelling estimates doxy-PEP could add ~4.4 million defined daily doses of doxycycline per year in the EU/EEA, raising **antimicrobial-resistance concerns** (PMID: 42565264).
- **Secondary prevention:** screening of asymptomatic high-risk individuals (multisite NAAT) with LGV genotyping of positives; treatment guideline publication was temporally associated with peaks in testing/diagnosis (PMID: 32762828).

- **Tertiary prevention:** prompt full-course therapy to prevent fibrotic complications.
- **Immunization:** no vaccine exists for LGV/*C. trachomatis*.

14. Other Species / Natural Disease

- **Taxonomy of the pathogen:** *Chlamydia trachomatis* (NCBI Taxon 813). Humans are the natural host; LGV is a **human-specific** disease with no established natural animal reservoir.
- **Zoonotic potential:** none established for the L-biovar.
- **Comparative biology:** related *Chlamydia* species cause disease in animals (e.g., *C. muridarum* in mice, *C. suis* in pigs, *C. abortus/psittaci* in livestock/birds), providing comparative models but not natural LGV.
- **Cross-species susceptibility:** notably, the human serovar **L2 can infect mice directly** (unlike most human serovars), which underpins its use in laboratory models (see Section 15).

15. Model Organisms

Mouse models of LGV use the human serovar L2 directly:

Model	System	Application	Evidence
Lung infection	C57BL/6J, serovars D and L2	Antibiotic (tetracycline, azithromycin) and vaccine screening; survival, bacterial load, histology, MPO, IFN- γ , TNF- α , MCP-1, IL-6	"we established an optimized lung infection model for the human intracellular bacterium <i>C. trachomatis</i> serovar D (and L2) in immunocompetent C57BL/6J mice" (PMID: 26676260)
Genital immunization	Attenuated plasmidless L2(25667R), intravaginal	Vaccine immunogenicity; partial protection	"Intravaginal immunization induced both chlamydial specific serum antibody and systemic CD4(+) Th1 biased immune responses" (PMID: 20004265)
Cross-protection	MoPn/human biovar challenge	Heterotypic immunity	Prior infection generates broadly cross-reactive T cells protecting against L2 challenge (PMID: 10338514)
Serovar comparison	Female upper genital tract	Immunopathology	"Infection with serovar D induces severe tissue inflammation in the female upper genital tract, whereas infection with serovar L2 does not" (PMID: 42413199)

Recapitulation/limitations: these models capture chlamydial replication, Th1/IFN- γ immunity, and antibiotic/vaccine responses, but the murine genital/lung models do **not** fully reproduce the human tertiary lymphatic fibrosis (buboes, strictures, elephantiasis) that defines LGV. In vitro models — HeLa/epithelial infection, monocyte-derived macrophage cultures, and T3SS-effector mutants — resolve the macrophage-tropism and invasion mechanisms. **Model databases:** MGI (mouse), Cellosaurus (HeLa, THP-1 cell lines).

Mechanistic Model / Interpretation

The unifying theme across all 15 findings is that **LGV is a pathogen-determined disease** — its distinctiveness among chlamydial infections flows entirely from the biology of the L-biovar, not from any host predisposition. Two pathogen properties convert a superficial mucosal infection into an invasive, fibrosing systemic disease:

1. **Macrophage tropism** (Finding 3): serovar L2 replicates productively in monocyte-derived macrophages, whereas non-disseminating urogenital serovars abort in these cells. This single biological difference provides the vehicle for lymphatic dissemination and is the mechanistic root of the "invasive" phenotype.
2. **A pro-dissemination inflammatory program** (Findings 3, 15): differential induction of IL-8, IL-11, TNF- α , and IDO shapes a recruitment and effector environment distinct from that of non-invasive strains.

Upstream of both sits the **T3SS-driven obligate intracellular developmental cycle** (Finding 13) common to all chlamydiae, which enables epithelial invasion and immune subversion. Downstream, the **IFN- γ /Th1 response controls the organism but, when chronic, together with MMP-mediated matrix remodeling, drives the scarring** (Finding 15) that produces the tertiary strictures, fistulae, and elephantiasis. The scarring paradigm is extrapolated from trachoma — the best-characterized chlamydial fibrosing disease — and represents the least directly demonstrated (but most biologically coherent) link in the LGV chain.

Clinically, this chain explains everything the surveillance and treatment data show: the anorectal-predominant proctitis of the MSM epidemic (Findings 2, 6, 10) is the mucosal expression; buboes are the lymphatic expression; and both are highly curable with a 21-day doxycycline course (Findings 5, 6, 14) precisely because antibiotics interrupt the cycle before irreversible fibrosis. Diagnostic delay (Finding 2) is the principal modifiable determinant of bad outcomes because it allows the chronic-inflammatory fibrotic arm to progress and because LGV proctitis masquerades as IBD or malignancy (Findings 2, 12).

Evidence Base

PMID	Role in report	What it supports
24518282	Foundational review	Etiology, 3-stage course, re-emergence, anorectal syndrome, NAAT diagnosis
39053939	Multicentre genetic study (Spain)	MSM 93.8%, HIV $\geq$ 43.5%, proctitis 73.3%
31243838	2019 European guideline	Strain distribution, 25% asymptomatic, 1:15 ratio, doxycycline regimen, contact treatment
27513890	Systematic review/meta-analysis	98.5% doxycycline cure rate
32762828	UK surveillance	Diagnosis trends, positivity decline, HIV share
22517888	Finnish genotyping study	Two-step NAAT-then-pmpH/ompA diagnosis
42151834	Polish case series	6-month delay, IBD/malignancy mimicry, persistent symptoms, 21-day doxycycline
3759241	In vitro	L2 macrophage tropism
11207588	In vitro	Differential IL-8/IL-11 cytokines
12011019	In vitro	Differential TNF- α /IDO signaling
33512479	Molecular review	T3SS, inclusion, developmental cycle
23457650	Trachoma immunology review	IFN- γ /Th1 protection; MMP-mediated scarring (extrapolated)
19460133	Genomics	Distinct invasive biovar; plasmid virulence
39915233	Reference-lab surveillance	Genovar evolution, emerging L1-like variant
16721218	Clinical/pathology	IBD-mimicking histopathology; prompt therapy response
7756478	Historical cluster (L1)	Serovar-dependent severity (L1 milder than L2)
34057249	2021 European proctitis guideline	Proctitis symptoms; asymptomatic fraction; condom limits
26676260	Mouse model	L2 lung model for antibiotic/vaccine screening

PMID	Role in report	What it supports
20004265	Mouse vaccine model	Attenuated L2, Th1 immunity, partial protection
10338514	Mouse cross-protection	Broadly cross-reactive T cells vs L2
42413199	Mouse comparison	Serovar-specific immunopathology (D vs L2)
42565264	Modelling	Doxy-PEP efficacy; AMR concern
42538441	Review	Doxy-PEP reduces chlamydia incidence

Evidence source types span **human clinical** (reviews, guidelines, case series, surveillance), **model organism** (murine L2 infection/vaccine studies), and **in vitro** (macrophage tropism, cytokine profiling, T3SS-effector mutants). No computational/structural predictions were required.

Limitations and Knowledge Gaps

- 1. Tertiary fibrosis mechanism is extrapolated, not directly demonstrated for LGV.** The IFN- γ /Th1 + MMP scarring paradigm (Finding 15) is drawn from trachoma; LGV-specific molecular studies of stricture/fistula formation are lacking.
- 2. No human host-genetics data.** Because LGV is infectious, sections on causal genes, variants, inheritance, penetrance, epigenetics, and chromosomal abnormalities are genuinely not applicable — but this also means host susceptibility modifiers (e.g., HIV-related immune status) are underexplored mechanistically.
- 3. Quality-of-life data are anecdotal.** No validated EQ-5D/SF-36/PROMIS measurements for LGV proctitis or tertiary disease were found.
- 4. Serovar–severity relationship is based on small/historical data.** The L1-milder-than-L2 claim rests on a single 1980s cluster ([PMID: 7756478](#)); the newly emerging L1-like variant's virulence is unknown.
- 5. Incidence/prevalence rates in per-100,000 terms are not well established** — LGV is reported as counts within MSM sexual-health surveillance rather than population rates.
- 6. Doxy-PEP evidence for LGV specifically** is inferred from chlamydia-wide efficacy; direct LGV-endpoint trials and AMR surveillance are still maturing.

7. **Antimicrobial resistance in *C. trachomatis*** to tetracyclines is currently not a documented clinical problem, but doxy-PEP scale-up warrants monitoring.

Proposed Follow-up Experiments / Actions

1. **Define the LGV fibrosis mechanism directly.** Use human anorectal biopsy transcriptomics/spatial transcriptomics (GEO) and MMP profiling from LGV proctitis/stricture tissue to test whether the trachoma MMP/Th1 model holds for LGV.
 2. **Characterize the emerging L1-like variant** ([PMID: 39915233](#)) — whole-genome sequencing, macrophage-tropism assays, and cytokine profiling versus L2b to assess virulence.
 3. **Establish validated QoL outcomes.** Apply EQ-5D/PROMIS instruments prospectively in symptomatic LGV proctitis and tertiary cohorts.
 4. **Longitudinal doxy-PEP surveillance** for LGV-specific incidence and tetracycline-resistance markers in *C. trachomatis*, given projected ~4.4M DDD/year increase in the EU/EEA ([PMID: 42565264](#)).
 5. **Reduce diagnostic delay** by embedding reflex LGV genotyping of all rectal *C. trachomatis*-positive NAATs in MSM and by clinician education on the IBD/malignancy mimicry (median 6-month delay; [PMID: 42151834](#)).
 6. **Develop an animal model of tertiary LGV** that recapitulates lymphatic fibrosis, since current murine genital/lung L2 models do not capture strictures/elephantiasis.
 7. **Vaccine research:** leverage the attenuated plasmidless L2(25667R) Th1 platform ([PMID: 20004265](#)) toward a broadly protective *C. trachomatis* vaccine, currently the major unmet prevention need.
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Report generated from 15 evidence-backed findings and 37 reviewed papers across 5 investigation iterations. All mechanistic and clinical claims are attributed to primary literature by PMID with verbatim abstract quotes.
